

Colour Evolution of GAMA Galaxies: Evidence for Halo-Mass-Mediated Quenching from Local Environment Measures

K. Sloan — Independent Researcher | Target: MNRAS Letters

Abstract

We investigate the relationship between local environment and galaxy colour ($u-r$) for 22,531 galaxies from the Galaxy and Mass Assembly (GAMA) DR4 survey at $0.002 < z < 0.35$. Using four environment measures — projected neighbour count, distance to fifth nearest neighbour, surface density, and adaptive density — we apply partial correlation analysis, residualising on halo mass ($\log M_A$), to isolate the net environmental signal. For the full galaxy population, environment correlates significantly with colour (Spearman $r = 0.18-0.25$, $p < 10^{-150}$), with approximately half the signal (45–58%) surviving after controlling for halo mass. For morphologically classified ellipticals ($N = 1,270$), the partial correlations collapse to $r \approx 0$ ($p > 0.3$), indicating that halo mass fully mediates the environment–colour relationship in this population. We additionally detect a significant monotonic trend in the shape of the $u-r$ colour distribution with redshift: the Box–Cox power λ increases from -0.20 at $z \approx 0.03$ to $+2.70$ at $z \approx 0.33$ (Spearman $r = +1.000$, $p = 4.0 \times 10^{-4}$), corresponding to a transition from a negatively skewed to a positively skewed distribution. A bootstrap null test confirms this trend is inconsistent with random sampling. These results support a picture in which galaxy quenching is primarily driven by halo mass, with the residual environment signal in the general population reflecting the halo mass–environment correlation rather than an independent environmental process.



Box–Cox λ as a Robust Shape Statistic for Asymmetric Distributions: A Comparison with Johnson S_U

K. Sloan — Independent Researcher | Target: MNRAS / A&A Methods

Abstract

Distribution shape statistics are widely used in astrophysics to characterise galaxy property distributions, assess normality, and detect evolutionary trends. We compare two approaches: the Johnson S_U (JSU) distribution, which fits four parameters (location, scale, shape a , shape b) simultaneously, and the Box–Cox power transform λ , which characterises distributional shape via a single, interpretable parameter. Using $u-r$ colour data from 165,488 GAMA DR4 galaxies divided into seven redshift bins ($0.002 < z < 0.35$), we demonstrate that the JSU shape parameters a and b exhibit large, non-monotonic jumps between adjacent bins (e.g., $a = -11.68$ at $z \approx 0.075$ to $a = +20.68$ at $z \approx 0.125$), arising from a correlated likelihood ridge in (a, b) parameter space. By contrast, the Box–Cox λ varies smoothly and monotonically from -0.20 to $+2.70$, enabling direct detection of a statistically significant shape evolution with redshift (Spearman $r = +1.000$, Kendall $\tau = +1.000$, $p = 4.0 \times 10^{-4}$). A bootstrap null test confirms this trend is inconsistent with random sampling ($p < 0.002$). We recommend λ as the preferred shape statistic when monotonic trends are of scientific interest, reserving JSU for applications requiring full distributional characterisation with stable parameter initialisation.

Guidelines for choosing between the two approaches are provided.